Incremental LEFM Prediction of a Crack Trajectory from a Circular Hole

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Abstract

Predicting a crack trajectory from an initially uncracked circular hole requires linking the initiation state to a sequence of mixed-mode crack-tip solutions. We consider this problem within quasi-static linear elastic fracture mechanics. The initiation point is selected by the maximum tensile tangential stress on the hole boundary, and the first segment follows the outward material-side normal. Subsequent directions are determined by the maximum tangential stress criterion using interaction-integral stress-intensity factors evaluated for the complete current crack geometry. Prescribed-field checks, crack-opening-displacement comparisons, and a centered-hole symmetry control assess the extraction and direction-selection procedures. Independently propagated exterior-mesh and structured-tip calculations test the sensitivity of the resulting trajectory. In the asymmetric case, the signed mode-mixity ratio reaches a positive maximum and then changes sign, reversing the incremental turning sense while the opening-mode intensity continues to increase under fixed remote traction. Independently propagated 4-, 2-, and 1-mm paths, refined together with their numerical length scales, exhibit contracting geometric and directional differences and nearly constant increment-normalized turning measures. The characteristic turning sequence persists under the tested mesh changes, while coupled refinement clarifies the dependence of mode mixity and turning density on the extension increment.

Keywords: crack trajectory; circular hole; linear elastic fracture mechanics; maximum tangential stress; mode mixity; interaction integral; crack-opening displacement.

1 Introduction

The initiation and subsequent propagation of a crack at a circular hole are related but distinct fracture-mechanics problems. Stress-intensity factors (SIFs) for prescribed radial and inclined hole-edge cracks have been studied since the classical work of Bowie [1]; mixed-mode solutions for finite plates [2] and weight-function methods [3] have extended their evaluation to other geometries and loading conditions. For an initially uncracked hole, the initiation site must first be determined from the stress field or a fracture criterion. Finite-fracture-mechanics analyses have examined both symmetric and asymmetric initiation [4] and subsequent growth stability under biaxial loading [5]. Selection of the initiation site, however, does not determine the path followed by a crack of finite length.

Once the first segment has formed, each change in crack direction alters the geometry and hence the mixed-mode field at the advancing tip. The maximum tangential stress (MTS) criterion of Erdogan and Sih [6] supplies a local direction rule in terms of the opening- and sliding-mode stress-intensity factors, K_{I} and K_{II}, respectively. Such criteria have long been incorporated into finite-element procedures with successive remeshing; Bouchard et al. [7], for example, compared alternative mixed-mode criteria in an automatic propagation framework. A cohesive-zone description instead resolves the fracture process zone rather than retaining an ideally sharp singular tip.

Recent potential-based cohesive calculations [8] provide a useful comparison with classical elastic kink predictions, although the present study is confined to linear elastic fracture mechanics (LEFM) and direction selection for prescribed crack extensions.

Hole-induced crack deflection has also been investigated using several computational approaches. Wang et al. [9] studied curved cracks emanating from adjacent holes; boundary-element methods have been applied to multiple cracks in perforated solids [10]. Fang et al. [11] combined adaptive extended isogeometric analysis with an interaction integral for cracks interacting with holes and voids, and compared trajectories obtained with two extension increments. Related studies include incremental remeshing for cracks from blasting holes [12], boundary-cracklet propagation in perforated plates [13], and finite-element simulations of hole-induced deflection [14]. These investigations demonstrate the range of available methods, but visual agreement between trajectories alone does not quantify the sensitivity of their successive direction updates.

The latter question is particularly important because numerical stress-intensity factors are not merely outputs of an incremental calculation: they determine the next geometry and therefore influence all subsequent solutions. Interaction integrals offer a means of separating the mixed-mode contributions [15], while displacement-extrapolation estimates are sensitive to near-tip element arrangements [16]. Comparisons of extraction procedures have emphasized the advantages of integral methods [17] and their dependence on adequate finite-element resolution [18]. Mixed-mode assessments have likewise examined the response of interaction-integral estimates to discretization choices [19]. For a recursively generated path, such local extraction differences can also alter the later crack geometries. Accuracy checks for a prescribed tip must therefore be distinguished from comparisons of independently propagated trajectories.

Here we examine this distinction for a single crack initiating at an asymmetric circular hole in a finite plate. The maximum tensile tangential stress of the uncracked-hole solution determines the initiation point and its outward material-side normal. Only the first crack segment is prescribed along that normal. At each subsequent step, the current elastic boundary-value problem is solved for the complete accumulated crack geometry, and the next direction is selected from the current-tip SIFs using the MTS criterion. The objective is not to introduce a new fracture criterion, but to establish which features of the resulting numerical trajectory remain stable under the investigated changes in extraction and discretization.

The assessment combines prescribed-field and fixed-geometry checks with independent crack-path calculations. Interaction-integral SIFs are compared with crack-opening-displacement (COD) extrapolations from the same computed displacement field; a centered-hole control probes reflection symmetry and the local one-step angular response. Independently propagated paths then test changes in exterior-mesh density and structured-tip resolution. A separate family with 4-, 2-, and 1-mm increments also refines the numerical length scales, so that it examines coupled increment/discretization sensitivity rather than the increment in isolation. We examine the positive maximum and subsequent sign reversal of mode mixity, their effect on the MTS turning direction, and the behavior of increment-normalized turning measures under coupled refinement.

Sections 2 and 3 define the physical problem and the incremental MTS construction. Sections 4 and 5 present the numerical assessment and crack-trajectory results, followed by the physical interpretation, limitations, and conclusions in Sections 6 and 7.

2 Problem formulation

2.1 Geometry, material, and loading

Figure 1 introduces the plate geometry, the coordinate conventions, and the applied loading. The specimen has width $2A$ and height B, with a circular hole of radius R whose center is denoted in the schematic by (x_c, y_c), where x_c and y_c are the coordinates measured from the lower-left corner. The computational Cartesian frame has its origin at the midpoint of the left edge, with

x pointing right and y upward. Its rectangular domain is

$$\Omega_{\text{rect}} = [0, 2A] \times [-B/2, B/2].$$

In this frame the hole center is $(x_c, y_c - B/2)$; no separate symbol for the shifted center ordinate is needed. The curved line emerging from the hole illustrates an admissible crack path; it is schematic and does not prescribe the computed trajectory. The initiation point P_0 is selected from the uncracked-hole solution, and only the first crack direction is prescribed. Subsequent directions follow from the current-tip SIFs.

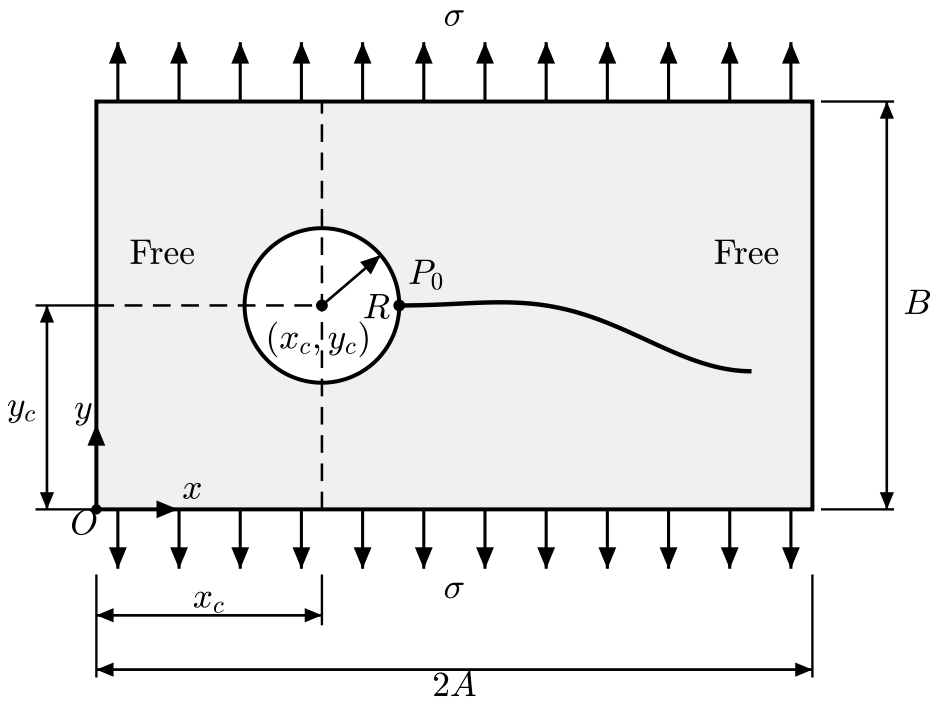


Figure 1: Plate geometry and tensile loading. Hole-center coordinates (x_c, y_c) are measured from the lower-left corner; the computational ordinate is shifted by $-B/2$. The horizontal edges carry outward tensile tractions of magnitude σ, while the remaining boundaries and crack faces are traction-free. The curved crack is schematic; dimensions are not to scale.

At step k, Ω_k is the material domain outside the hole and the retained crack Γ_k, with distinct upper and lower crack faces. The material is homogeneous, isotropic, and linearly elastic, with Young's modulus E and Poisson's ratio ν, under plane strain. For the crack geometry Γ_k ending at P_k, the displacement field $\boldsymbol{u}$, strain tensor $\boldsymbol{\varepsilon}$, and stress tensor $\boldsymbol{\sigma}$ satisfy

$$\nabla \cdot \boldsymbol{\sigma} = \mathbf{0} \qquad \text{in } \Omega_k, \tag{1}$$

$$\boldsymbol{\varepsilon} = \tfrac{1}{2}(\nabla \boldsymbol{u} + \nabla \boldsymbol{u}^{\mathsf{T}}), \qquad \varepsilon_{33} = 0, \tag{2}$$

$$\boldsymbol{\sigma} = 2\mu\boldsymbol{\varepsilon} + \lambda \operatorname{tr}(\boldsymbol{\varepsilon})\boldsymbol{I}, \quad \mu = \frac{E}{2(1+\nu)}, \quad \lambda = \frac{E\nu}{(1+\nu)(1-2\nu)}. \tag{3}$$

Here $\boldsymbol{I}$ is the identity tensor, and μ and λ are the shear modulus and first Lamé coefficient, both with stress units. Uniform outward tensile tractions act on the horizontal edges, in the directions indicated by the arrows in Fig. 1. Their unsigned magnitude is $\sigma > 0$, as in the drawing. The traction vectors on the upper and lower edges are $(0, \sigma)$ and $(0, -\sigma)$, respectively. The vertical sides, hole boundary, and crack faces are traction free. Rigid-body motions are removed by imposing $u_x = u_y = 0$ at the lower-left corner and $u_y = 0$ at the lower-right corner. There are no body forces.

For fixed geometry, positive scaling of σ scales both SIFs equally, preserving their ratio and the MTS direction. The present formulation predicts the direction of imposed quasi-static increments, not a load-versus-time crack-growth history.

2.2 Stress-selected initiation and the frozen frame

The uncracked-hole elastic field supplies the tangential boundary stress $\sigma_{tt}(\phi)$, where ϕ is measured counterclockwise from the positive Cartesian x direction. Its maximum tensile value selects the initiation point; the numerical boundary evaluation is described in Section 4. Let σ_{ini} denote the prescribed tensile-stress threshold for initiation, not a cohesive strength. The dimensionless initiation load multiplier Λ_{ini} is the ratio of this threshold to the maximum tensile stress under the reference loading; it is distinct from the first Lamé coefficient λ. The initiation rule is

$$\phi_* = \arg\max_{\phi} \sigma_{tt}^{+}(\phi;\sigma), \qquad \Lambda_{\mathrm{ini}} = \frac{\sigma_{\mathrm{ini}}}{\max_{\phi} \sigma_{tt}^{+}(\phi;\sigma)}, \quad \sigma_{tt}^{+} = \max(\sigma_{tt}, 0). \tag{4}$$

The fitted angle ϕ_* specifies the initiation point and the outward material-side radial normal:

$$P_0 = \begin{bmatrix} x_{\mathrm{c}} + R\cos\phi_* \\ y_{\mathrm{c}} - B/2 + R\sin\phi_* \end{bmatrix}. \tag{5}$$

The stress-based initiation condition supplies the location and the corresponding load multiplier; it is not reused as a growth or arrest criterion along the crack. No cohesive traction–separation law is introduced in the present LEFM formulation.

The frozen basis is

$$\boldsymbol{n}_{\mathrm{mat}} = \begin{bmatrix} \cos\phi_* \\ \sin\phi_* \end{bmatrix}, \qquad \boldsymbol{t} = \begin{bmatrix} -n_{\mathrm{mat},y} \\ n_{\mathrm{mat},x} \end{bmatrix}. \tag{6}$$

The normal points radially out of the hole and into the material; $\boldsymbol{t}$ is its counterclockwise rotation. All absolute crack angles θ_k below are measured relative to this frozen normal, not to the global x axis. Its global orientation is $\phi_* + \theta_k$; translating the coordinate origin does not change this angle. With increment length Δa, the initial direction and first segment are prescribed by

$$\theta_1 = 0, \qquad P_1 = P_0 + \Delta a\, \boldsymbol{n}_{\mathrm{mat}}. \tag{7}$$

The initiation point and frame remain frozen for subsequent increments. The study treats one active crack; competing subsequent initiation sites are outside its scope.

3 Incremental crack-path formulation

3.1 Current-tip frame and stress-intensity factors

At the current tip P_k, the local forward unit vector $\boldsymbol{e}_1^{(k)} = (P_k - P_{k-1})/\Delta a$ follows the last crack segment; $\boldsymbol{e}_2^{(k)} = (-e_{1y}^{(k)}, e_{1x}^{(k)})$ is its counterclockwise normal. Thus the local frame follows the current tip direction rather than the chord connecting the crack mouth to the tip. With x_1 and x_2 measured along these vectors and $r > 0$ measured directly ahead of the tip, the opening- and sliding-mode stress-intensity factors are defined by

$$K_{\mathrm{I}}^{(k)} = \lim_{r\to 0^+} \sqrt{2\pi r}\, \sigma_{22}(r,0), \qquad K_{\mathrm{II}}^{(k)} = \lim_{r\to 0^+} \sqrt{2\pi r}\, \sigma_{12}(r,0). \tag{8}$$

Here σ_{22} and σ_{12} are the stress components in the current-tip frame. The signed mode-mixity ratio is defined for positive opening-mode intensity by

$$q_{\mathrm{K}}^{(k)} = \frac{K_{\mathrm{II}}^{(k)}}{K_{\mathrm{I}}^{(k)}}, \qquad K_{\mathrm{I}}^{(k)} > 0. \tag{9}$$

This dimensionless ratio has the sign of the sliding-mode intensity. Both stress-intensity factors are extracted from an elastic solution that retains the complete preceding crack geometry.

3.2 The implemented MTS rule

The maximum tangential stress (MTS) criterion determines the direction of a new segment relative to the preceding one. Let α be the counterclockwise turning angle in local polar coordinates centered at the crack tip, and write $c = \cos(\alpha/2)$ and $s = \sin(\alpha/2)$. The functions $S(\alpha)$ and $T(\alpha)$ are the angular amplitudes of the leading circumferential normal stress and radial–circumferential shear stress, respectively:

$$S(\alpha) = K_{\mathrm{I}}^{(k)} c^3 - 3K_{\mathrm{II}}^{(k)} sc^2, \tag{10}$$

$$T(\alpha) = K_{\mathrm{I}}^{(k)} sc^2 + K_{\mathrm{II}}^{(k)} c(1 - 3s^2). \tag{11}$$

Both stresses have the common singular factor $1/\sqrt{2\pi r}$. Since $S'(\alpha) = -3T(\alpha)/2$, the stationary circumferential-stress directions satisfy $T(\alpha) = 0$. The numerical procedure selects the tensile root continuously connected to straight extension as $K_{\mathrm{II}}^{(k)} \to 0$. The accepted states have positive opening intensity and small turning angles; the implemented root search and its safeguards are unchanged.

For $|q_{\mathrm{K}}^{(k)}| \ll 1$, expansion on this branch gives

$$\alpha_{\mathrm{MTS}} = -2q_{\mathrm{K}}^{(k)} + \frac{14}{3}(q_{\mathrm{K}}^{(k)})^3 + O((q_{\mathrm{K}}^{(k)})^5), \tag{12}$$

where the angle is in radians. In particular, positive mode mixity produces clockwise turning in the adopted frame. This expansion is used for interpretation, whereas the reported trajectories use the unapproximated MTS root.

3.3 Finite-increment recurrence

The unit direction of a segment is expressed in the frozen initiation basis. With Δa denoting the prescribed segment length, the incremental turning angle, the absolute direction of the new segment, and its tip position satisfy

$$\boldsymbol{e}(\theta) = \cos\theta\, \boldsymbol{n}_{\mathrm{mat}} + \sin\theta\, \boldsymbol{t}, \tag{13}$$

$$\Delta\theta_{k+1} = \alpha_{\mathrm{MTS}}(K_{\mathrm{I}}^{(k)}, K_{\mathrm{II}}^{(k)}), \tag{14}$$

$$\theta_{k+1} = \theta_k + \Delta\theta_{k+1}, \tag{15}$$

$$P_{k+1} = P_k + \Delta a\, \boldsymbol{e}(\theta_{k+1}). \tag{16}$$

Here θ_k is measured relative to the fixed outward normal at the initiation point. The accumulated crack length is $a_k = k\Delta a$; throughout the numerical results, a denotes this length rather than the horizontal tip coordinate. The accepted elastic solution for each current geometry determines the following segment, so the propagation history enters recursively through the boundary-value problem.

The turning density $\Delta\theta_{k+1}/\Delta a$ is the signed angle change per unit extension. It does not represent the pointwise curvature of the polygonal crack, whose direction is discontinuous at its vertices. Consider instead a family of decreasing increments and suppose that the computed polygons approach an arc-length-parametrized curve $\boldsymbol{r}(a)$ with a differentiable tangent angle $\theta(a)$. If, in addition, their discrete turning densities converge at corresponding crack lengths to the signed curvature $\mathrm{d}\theta/\mathrm{d}a$, the small-turn MTS expansion implies

$$\frac{q_{\mathrm{K}}^{(k)}}{\Delta a} = -\frac{1}{2}\frac{\Delta\theta_{k+1}}{\Delta a} + O((\Delta a)^2) \longrightarrow -\frac{1}{2}\frac{\mathrm{d}\theta}{\mathrm{d}a}. \tag{17}$$

Indeed, a bounded turning density implies $\Delta\theta_{k+1} = O(\Delta a)$, and therefore $q_{\mathrm{K}}^{(k)} = O(\Delta a)$ on the selected small-turn branch. Positive curvature corresponds to counterclockwise turning, hence

to negative mode mixity. This relation is a conditional smooth-path interpretation, not a convergence theorem: convergence of polygonal positions alone does not ensure convergence of the turning densities, and numerical SIF extraction must resolve mode mixity at the scale of Δa.

4 Numerical implementation and verification

The computation begins with stress-selected initiation and proceeds by solving successive cracked geometries with a structured near-tip mesh. We describe the interaction-integral and crack-opening-displacement (COD) extractions, the numerical acceptance criteria, and the mesh configurations used to assess their sensitivity. The subsequent tests address two distinct questions. Prescribed fields and fixed crack geometries examine SIF extraction without changing the path, whereas independently propagated calculations reveal how differences in the discretization affect successive crack directions. A coupled 4/2/1-mm refinement study and a centered-hole symmetry control complete the numerical assessment. The trajectory results are presented in Section 5.

4.1 Numerical parameters and initiation state

For the computations, $2A = 300$ mm, $B = 200$ mm, and $R = 30$ mm. The center is $(x_c, y_c) = (170, 80)$ mm in the lower-left convention of Fig. 1; in the midheight computational frame it is $(170, -20)$ mm. The material constants are $E = 210$ GPa and $\nu = 0.30$ (plane strain). The reference tensile traction is $\sigma = 1$ MPa, and the tensile initiation threshold is $\sigma_{\text{ini}} = 300$ MPa.

The uncracked-hole calculation (Stage I) uses a 480-segment boundary polygon containing the fitted initiation point. Tangential stresses obtained from six-node quadratic triangles (T6) are evaluated at three material-side offsets and extrapolated linearly to the hole boundary. A local quadratic angular fit then gives $\phi_* = -1.5606^\circ$ and $P_0 = (0.199989, -0.0208170)$ m. Under the unit reference traction, the peak tangential stress is 3.7141 MPa and the initiation multiplier is $\Lambda_{\text{ini}} = 80.774$. The corresponding frozen radial normal is $(0.999629, -0.0272345)$. These Stage-I quantities are retained throughout the propagation studies. The reference path uses $\Delta a = 4$ mm, whereas the increment-refinement family also includes 2- and 1-mm steps. Unless otherwise indicated, SIFs are reported for unit reference traction in MPa$\sqrt{\text{m}}$.

4.2 Geometry and paired tip discretization

At each increment the finite-element geometry contains the complete crack polyline, including previous kinks. During mesh generation, a narrow auxiliary notch (meshing channel) is merged with the circular hole to represent the two crack faces. After meshing, the opposing notch faces are collapsed onto the crack polyline while retaining distinct upper and lower nodal identities. This approach is related to the auxiliary meshing channel used in the cohesive crack-kinking formulation of Bogdanov and Selivanov [8]. Here the collapsed faces are traction-free; no cohesive traction–separation law is introduced.

The current-tip region has a structured mesh with characteristic spacing h_{tip}, paired by reflection across the last crack segment. A conforming graded mesh resolves the remaining plate and its asymmetric boundaries. Three-node triangles (T3) are then converted to straight-sided quadratic T6 elements.

The inner and outer radii of the interaction-integral annulus are r_{in} and r_{out}, respectively; the radius of the structured tip mesh is r_{core}. The reference numerical scales are

$$\frac{h_{\text{tip}}}{\Delta a} = 0.00675308135, \quad \frac{r_{\text{in}}}{\Delta a} = 0.10, \quad \frac{r_{\text{out}}}{\Delta a} = 0.65, \quad \frac{r_{\text{core}}}{\Delta a} = 0.75. \tag{18}$$

For the 4-mm reference increment these give $h_{\text{tip}} = 0.027012$ mm, $r_{\text{in}} = 0.4$ mm, $r_{\text{out}} = 2.6$ mm, and $r_{\text{core}} = 3$ mm. The structured core contains 12678 T3 triangles, of which 11316 contribute to the interaction-integral evaluation.

Every candidate mesh is checked for retention of the physical boundary, separate crack faces, preservation of the intact ligament, and conforming interfaces between structured and exterior elements. Element areas and T6 Jacobians must be positive; the minimum triangle angle is 20°, and the maximum permitted adjacent longest-edge ratio is 1.8. These requirements establish geometric and discretization admissibility, not the accuracy of the elastic solution.

Two exterior mesh-sizing laws are considered. The reference law, denoted M0, uses a far-field element-size cap of 2.5 mm, a grading slope of 0.10, and a boundary-metric growth parameter of 0.25. The coarser law M1 increases these values to 5.0 mm, 0.15, and 0.35, respectively. Both laws retain the 4-mm grading transition length in the reference-increment comparisons. M0 is used for the reference 4-mm trajectory and the structured-tip resolution tests, including the independently propagated $2h_0$ path. M1 is used for the fixed-geometry exterior-mesh comparison and the independently propagated exterior-mesh trajectory. The coupled 4/2/1-mm refinement study, described below, uses dimensionless versions of the M1 exterior law and a coarser structured-tip family.

For the fixed P_{17} and P_{22} geometries, changing from M0 to M1 reduces the total T3 element count by approximately 31–33% and the exterior count by 40–43%. The paired 12678-element near-tip core and its 11316-element interaction-domain support remain unchanged, permitting a comparison of the extracted SIFs before the M1 trajectory is propagated independently.

4.3 Structured-tip resolution tests

The influence of structured-tip resolution is first examined on fixed accepted crack geometries. The mesh spacing near the current tip is varied while the complete crack polyline, material, loading, interaction-domain radii, far-field cap, transition length, and MTS criterion remain unchanged. The production tip size is denoted by $h_0 = 0.027012$ mm. The three physically compared resolutions are

$$2h_0 = 0.054025 \text{ mm}, \qquad h_0 = 0.027012 \text{ mm}, \qquad h_0/2 = 0.013506 \text{ mm}. \tag{19}$$

These correspond to structured-core scale factors 2, 1, and 1/2. The paired-core radius remains $r_{\text{core}} = 3$ mm and the EDI annulus remains $r_{\text{in}} = 0.4$ mm to $r_{\text{out}} = 2.6$ mm. The paired cores contain 3318, 12678, and 49518 T3 elements, respectively. COD fitting uses native crack-face node pairs, namely corresponding upper and lower T6 nodes at the same distance behind the crack tip. Their sampling adequacy is discussed with the extraction procedure in Section 4.5.

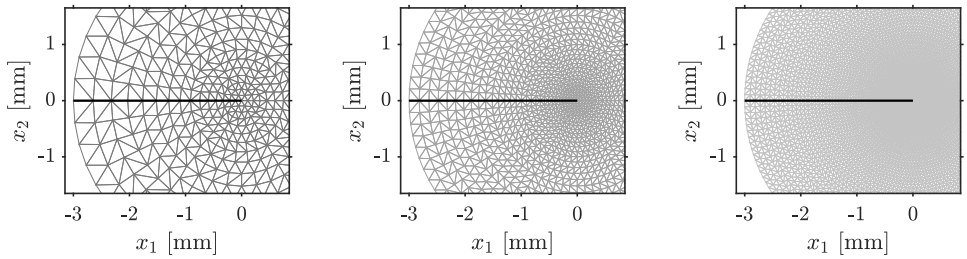
The levels $2h_0$, h_0, and $h_0/2$ differ only in the prescribed structured-tip scale and the conforming remeshing it requires. The M0 exterior sizing law remains fixed, although the exterior connectivity may change to conform to each core interface. This study consequently examines near-tip resolution, complementing the M0–M1 exterior-size comparison.

To examine accumulated directional sensitivity, the $2h_0$ discretization is also propagated through 92 mm of crack extension. The prescribed first segment P_0P_1 is common to the two paths; thereafter, each MTS update follows from the elastic field of the corresponding path. The $h_0/2$ level is used only in the fixed-geometry comparison.

For completeness, Fig. 2 also shows a $4h_0$ core ($h_{\text{tip}} = 0.10805$ mm) for geometric illustration. Its exclusion from physical SIF comparisons is explained by the COD sampling criterion in Section 4.5.

4.4 Interaction-integral extraction

The opening- and sliding-mode SIFs are obtained from the interaction integral in equivalent-domain form. Let w be its radial weight function and let m label the opening- or sliding-mode auxiliary field. The component indices $i, j \in \{1, 2\}$ refer to the current-tip frame and are summed when repeated; δ_{1j} is the Kronecker delta. For the actual stress, strain, and displacement fields



(a) $4h_0$ (b) $2h_0$ (c) h_0

Figure 2: Structured-tip meshes with a common core radius $r_{\text{core}} = 3$ mm. The reference spacing is $h_0 = 0.027012$ mm; the illustrative $4h_0$ level does not satisfy the COD sampling criterion.

$(\boldsymbol{\sigma}, \boldsymbol{\varepsilon}, \boldsymbol{u})$, let Ω_k denote the integration domain around the current tip. Here $\mathrm{d}\Omega$ is the area element, a comma followed by an index denotes partial differentiation in the local frame, and the colon denotes tensor contraction. The interaction integral reads

$$I^{(m)} = \int_{\Omega_k} \left[-(\boldsymbol{\sigma}^{(m)} : \boldsymbol{\varepsilon})\delta_{1j} + \sigma_{ij} u_{i,1}^{(m)} + \sigma_{ij}^{(m)} u_{i,1} \right] w_{,j} \, \mathrm{d}\Omega, \qquad i, j \in \{1, 2\}. \tag{20}$$

The gradients of the actual displacement field are evaluated with T6 shape functions, whereas the auxiliary stresses are taken from the asymptotic crack-tip fields. Derivatives of the auxiliary displacements are evaluated by centered finite differences with step $h = \max(10^{-7}\ \mathrm{m}, 10^{-5} r)$, where r is the distance from the current tip. The nodal weight w equals one for $r \le r_{\text{in}}$, decreases linearly to zero at r_{out}, and vanishes beyond the annulus. Its T6 interpolant provides the gradient used in Eq. (20); numerical integration employs the 16-point triangular rule. For unit auxiliary intensity $K_{\text{aux}} = 1$ in plane strain,

$$K_m = \frac{E'}{2K_{\text{aux}}} I^{(m)}, \qquad E' = \frac{E}{1 - \nu^2}, \qquad m \in \{I, II\}. \tag{21}$$

4.5 COD-based extraction comparison

Crack-opening displacement (COD) provides a second route to stress-intensity factors from the same computed elastic field. Each native crack-face node pair consists of matching upper and lower T6 nodes on the last segment, with both nodes located at distance r behind the current tip. The displacement jump $[\![u_i]\!]$ is the upper-minus-lower difference in component u_i at such a pair. These existing nodal positions are used directly; no additional interpolation points are introduced. The corresponding apparent COD SIFs are

$$\begin{bmatrix} K_{\text{I}}^{\text{app}}(r) \\ K_{\text{II}}^{\text{app}}(r) \end{bmatrix} = \frac{\mu}{\kappa + 1} \sqrt{\frac{2\pi}{r}} \begin{bmatrix} [\![u_2]\!] \\ [\![u_1]\!] \end{bmatrix}, \qquad \kappa = 3 - 4\nu. \tag{22}$$

Here κ is the dimensionless plane-strain Kolosov constant. Linear and quadratic fits of the apparent SIFs against the normalized distance $r/\Delta a$ are extrapolated to $r = 0$. The four fitting intervals are $[0.04, 0.20]$, $[0.04, 0.30]$, $[0.08, 0.30]$, and $[0.12, 0.30]$. The intervals overlap, so their sampling counts are not additive. In the stated order, the numbers of native crack-face node pairs are 19, 28, 23, and 18 for the $2h_0$ mesh; 38, 55, 44, and 34 for the reference h_0 mesh; and 74, 108, 86, and 67 for the $h_0/2$ mesh.

The prescribed qualification criterion requires at least 12 pairs in each fitting interval. The illustrative $4h_0$ core provides 10, 15, 12, and 9 pairs, respectively, and fails this criterion in the first and fourth intervals. It is therefore excluded from the physical SIF and MTS comparisons. Linear and quadratic fitting over the four intervals gives eight COD estimates from the same finite-element displacement field, not eight independent elastic solutions. Only the EDI SIFs determine subsequent MTS directions, and the differences among COD fits do not represent a statistical uncertainty interval.

4.6 Numerical acceptance and extraction checks

The assembled stiffness system is reduced to the free degrees of freedom and symmetrized as specified by the verified formulation. A symmetric approximate minimum-degree ordering is used together with symmetric Gauss–Seidel preconditioning and the preconditioned conjugate-gradient method (PCG). The reported-residual tolerance is 10^{-10}, the maximum iteration count is 5000, and the independently calculated true-relative-residual acceptance criterion is 5×10^{-10}. Prescribed displacement constraints are enforced exactly. For the finer fixed-geometry sensitivity meshes only, MATLAB PCG can report finite-precision stagnation immediately above the stated tolerance. In that case one Krylov restart from the returned iterate is permitted, using the same matrix, right-hand side, SGS factors, tolerance, and iteration limit. The final PCG and independently recomputed residual gates are unchanged; a restart therefore resets the recurrence but does not relax acceptance. SIF extraction and MTS direction selection are performed only on fields satisfying these residual criteria.

As a separate test of the extraction procedure, nodal displacements are prescribed from analytical pure-I, pure-II, and small mixed-mode asymptotic fields, with $(K_{\mathrm{I}}, K_{\mathrm{II}}) = (1, 10^{-4})$ in the mixed case. The allowed recovery and small mixed-mode relative errors are 2×10^{-4}; the limits for pure-mode cross-leakage and superposition are 10^{-10}. On the reference meshes, the largest observed small mixed-mode relative error is 3.86×10^{-8}. Because the fields are prescribed rather than obtained by solving the plate boundary-value problem, these tests qualify modal extraction and mesh topology but do not bound the error of the physical stress-intensity factors.

The reference path contains accepted physical solutions through P_{23}. The candidate extension to P_{24} passes geometric and prescribed-field qualification but has no accepted elastic solution. Across the accepted P_2–P_{23} solves, the largest true relative residual is about 1.00×10^{-10}, with 2442–3011 PCG iterations. Figure S1 summarizes these numerical acceptance diagnostics, whose full-precision values are retained in the saved evidence.

4.7 Crack-increment refinement family

The influence of the crack increment is examined using three independently propagated paths with

$$\Delta a \in \{4, 2, 1\} \text{ mm}.$$

These calculations use a common dimensionless coarse family, distinct from the reference 4-mm calculation with exterior law M0 and structured-tip spacing h_0. The local tip spacing is twice the reference dimensionless spacing, and the exterior follows the dimensionless M1 sizing law. Thus

$$\frac{h_{\mathrm{tip}}}{\Delta a} = 2(0.00675308135), \qquad \frac{r_{\mathrm{in}}}{\Delta a} = 0.10, \qquad \frac{r_{\mathrm{out}}}{\Delta a} = 0.65, \qquad \frac{r_{\mathrm{core}}}{\Delta a} = 0.75. \tag{23}$$

The exterior follows the M1 family, with maximum requested element size h_{far} and grading transition length L_{tr} given by

$$\frac{h_{\mathrm{far}}}{\Delta a} = 1.25, \qquad \frac{L_{\mathrm{tr}}}{\Delta a} = 1.00,$$

and with far-field grading slope 0.15 and boundary-metric growth 0.35. The absolute structured-tip spacing, core radius, interaction-domain radii, and exterior scales therefore decrease with Δa. This comparison tests a coupled increment/discretization family; it does not isolate the crack increment with the physical mesh fixed.

All three calculations retain the accepted Stage-I initiation point and frozen frame, but use their own first-segment lengths and independently solved P_1 states. Every subsequent vertex is generated from the corresponding path's own SIF/MTS history until its accumulated crack length reaches 92 mm. Corresponding-tip positions and directions are compared at $a = 4, 8, \ldots, 92$ mm, the exact common lengths of the three histories; no trajectory interpolation is involved.

Separately, linear interpolation between solved states of each history provides an estimate of the mode-mixity zero $q_{\mathrm{K}} = 0$ [Eq. (9)]. This is an event-location diagnostic, not an independently solved local-symmetry state.

4.8 Centered-hole symmetric control

To assess reflection symmetry and the local angular response of the numerical procedure, we consider a centered-hole plate with the same overall dimensions and hole radius. The hole center is $(0.150, 0)$ m, and the crack starts from its rightmost point,

$$P_0 = (0.180, 0)\ \mathrm{m}, \qquad P_1 = (0.184, 0)\ \mathrm{m},$$

with $\Delta a = 4$ mm. The straight crack is reflection symmetric and therefore has a pure mode-I field in the ideal continuum problem. The one-step angular response is probed by prescribing five two-segment crack geometries,

$$\theta_2 \in \{-0.10^\circ, -0.05^\circ, 0, +0.05^\circ, +0.10^\circ\},$$

and solving each geometry independently with the same tip discretization, EDI extraction, and elastic solver as in the asymmetric case. Let F denote the predicted direction of the third segment. The one-step MTS map is

$$F(\theta_2) = \theta_3 = \theta_2 + \alpha_{\mathrm{MTS}}(K_{\mathrm{I}}(\theta_2), K_{\mathrm{II}}(\theta_2))\,. \tag{24}$$

At the straight state,

$$K_{\mathrm{I}}(0) = 0.42541\ \mathrm{MPa}\sqrt{\mathrm{m}}, \qquad \frac{K_{\mathrm{II}}(0)}{K_{\mathrm{I}}(0)} = 6.22 \times 10^{-9},$$

and the predicted next direction is $F(0) = -7.13 \times 10^{-7}$ degrees, numerically indistinguishable from the symmetry line at the scale relevant here. The perturbed probes give the map summarized in Table 1.

Table 1: One-step directional map for the centered-hole symmetric control. The prescribed second-segment angle and the predicted third-segment direction are shown.

θ_2 [deg]	$F(\theta_2) = \theta_3$ [deg]
-0.10	$+0.011415$
-0.05	$+0.005708$
0	-7.13×10^{-7}
$+0.05$	-0.005708
$+0.10$	-0.011415

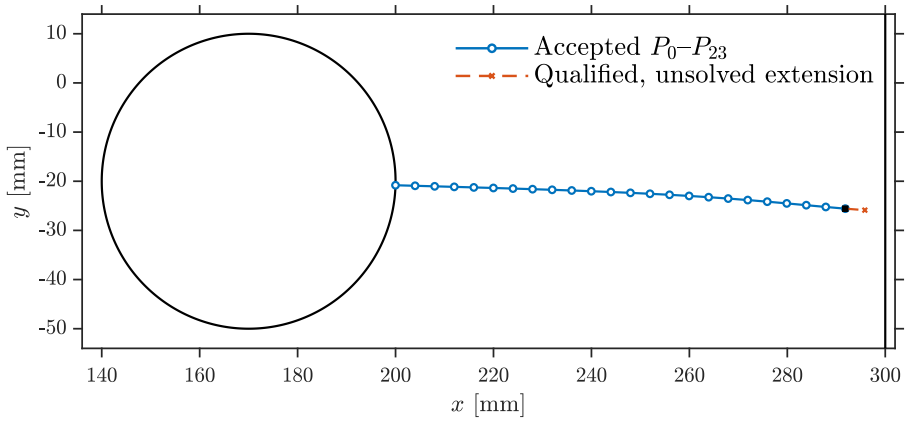
The opening-mode intensity is approximately even in the prescribed perturbation θ_2, whereas the signed mode-mixity ratio and $F(\theta_2)$ exhibit the expected approximately odd responses under reflection. The small residual at the straight configuration illustrates that these symmetry relations are recovered numerically rather than as exact identities. The two symmetric finite-difference estimates are

$$\frac{F(+0.05^\circ) - F(-0.05^\circ)}{0.10^\circ} = -0.11416, \qquad \frac{F(+0.10^\circ) - F(-0.10^\circ)}{0.20^\circ} = -0.11415,$$

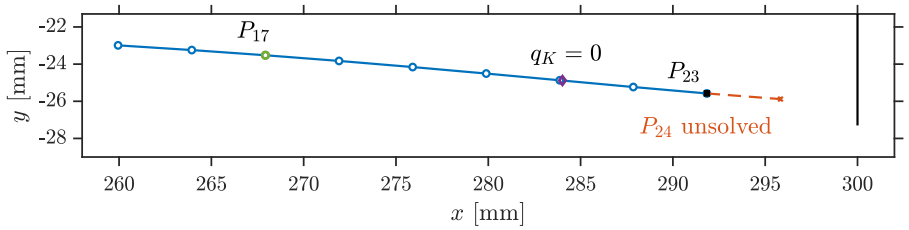
and a least-squares fit constrained through the origin yields the local slope estimate

$$F'(0) \approx -0.11415. \tag{25}$$

For these two-segment geometries at $\Delta a = 4$ mm, the next-segment angular response changes sign and is approximately 8.8 times smaller than the imposed perturbation. Because later steps change the complete crack geometry, they cannot be interpreted as repeated applications of the same scalar map F. The fitted slope therefore characterizes the local one-step response, not stability of the evolving trajectory.



(a) Accepted trajectory and proposed extension.



(b) Late-path detail and characteristic locations.

Figure 3: Crack trajectory from the asymmetric hole. Each panel has equal horizontal and vertical scales. The solid line ends at the last accepted tip P_{23}; the dashed $P_{23}P_{24}$ extension has no accepted elastic solution. The diamond marks the interpolated mode-mixity zero, not a solved crack tip.

5 Crack-trajectory results

5.1 Accepted path and direction history

The prescribed radial segment P_0P_1 establishes the first crack-tip configuration. Under the unit reference traction, its opening- and sliding-mode SIFs are $K_{\mathrm{I}}^{(1)} = 0.36648$ and $K_{\mathrm{II}}^{(1)} = 4.1983\times10^{-6}$ MPa$\sqrt{\mathrm{m}}$, respectively. Their small positive mode-mixity ratio gives an initial clockwise correction, $\Delta\theta_2 = \theta_2 = -0.0013127°$. Each subsequent accepted crack direction is obtained from a newly solved elastic field, producing the history through P_{23}.

Figure 3 shows the resulting P_0–P_{23} polyline, with a solid line reserved for accepted crack geometry. The total accepted length is 92 mm. Its last tip and remaining horizontal clearance from the right boundary are

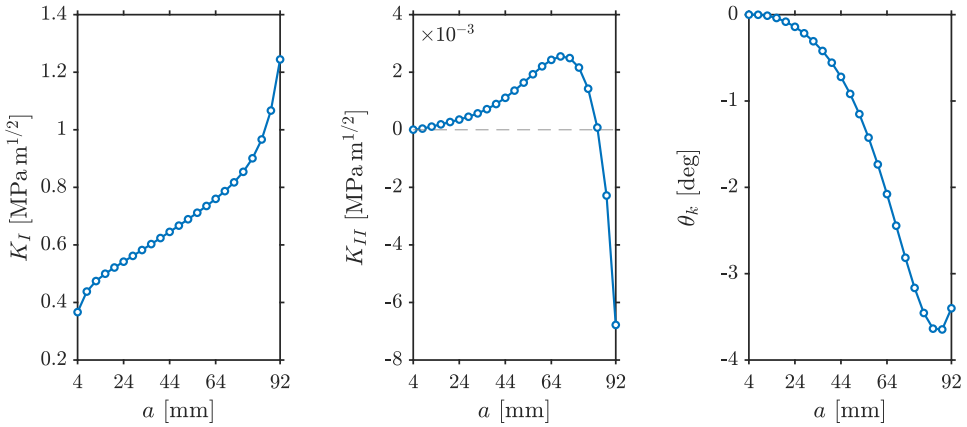
$$P_{23} = (291.840, -25.579)\ \mathrm{mm}, \qquad 2A - x_{23} = 8.16\ \mathrm{mm}. \tag{26}$$

The next candidate segment, $P_{23}P_{24}$, is shown dashed because its geometry is qualified but the elastic problem at P_{24} has not been solved to the acceptance criterion.

The polygonal path initially turns clockwise with increasing magnitude of its negative inclination. Its segment direction reaches $\theta_{22} = -3.646°$ and becomes less negative at P_{23} (Fig. 4). This change reflects a positive incremental turn, not a reversal of vertical motion: the final accepted segment still points below the horizontal. The trajectory is piecewise linear, and no continuously curved path has been solved independently.

5.2 Evolution of stress-intensity factors and mode mixity

The opening-mode intensity increases monotonically along the accepted 4-mm history, from $K_{\mathrm{I}} = 0.36648$ at P_1 to 1.2444 at P_{23} (Fig. 4). These values are evaluated under the same unit remote



(a) Opening-mode factor K_{I}. (b) Sliding-mode factor K_{II}. (c) Absolute direction θ_k.

Figure 4: Opening- and sliding-mode SIFs (MPa$\sqrt{\mathrm{m}}$) and segment direction (degrees) versus accumulated crack length a (mm), under unit remote traction. The segment direction reaches its minimum at P_{22} and becomes less negative at P_{23}.

traction, so the increase reflects the changing elastic geometry rather than an applied-load history.

The sliding-mode intensity begins close to zero and reaches its largest positive value, $K_{\mathrm{II}} = 0.0025484$, at P_{17} ($a = 68$ mm). The signed mode-mixity ratio defined in Eq. (9) attains its positive maximum, $q_{\mathrm{K}} = 0.0032389$, at the same state. This is a positive maximum, not a maximum of $|K_{\mathrm{II}}|$: the negative sliding-mode magnitude at the last accepted tip is larger. After P_{17}, K_{II} decreases while K_{I} continues to rise. The resulting decrease in positive mode mixity reduces the magnitude of the clockwise MTS turn through P_{21} (Fig. 5).

5.3 Mode-mixity sign change and turning reversal

The mode-mixity ratio remains positive at P_{21} ($a = 84$ mm), where $q_{\mathrm{K}}^{(21)} = +8.134267 \times 10^{-5}$, but is negative at P_{22} ($a = 88$ mm), where $q_{\mathrm{K}}^{(22)} = -0.0021416$. Thus the sign reversal is established between two independently solved accepted states, although neither has zero mode mixity. For the graphical marker in Figs. 3b and 5, linear interpolation between their values gives

$$a_{\mathrm{LS}} = a_{21} - \frac{q_{\mathrm{K}}^{(21)}}{q_{\mathrm{K}}^{(22)} - q_{\mathrm{K}}^{(21)}}(a_{22} - a_{21}) \simeq 84.15 \text{ mm}. \tag{27}$$

Here a_{LS} denotes only an interpolated estimate of the crack length at which local symmetry ($K_{\mathrm{II}} = 0$) would occur. It is not a solved zero-mode-II state, and the interpolation does not establish the physical location with sub-increment accuracy.

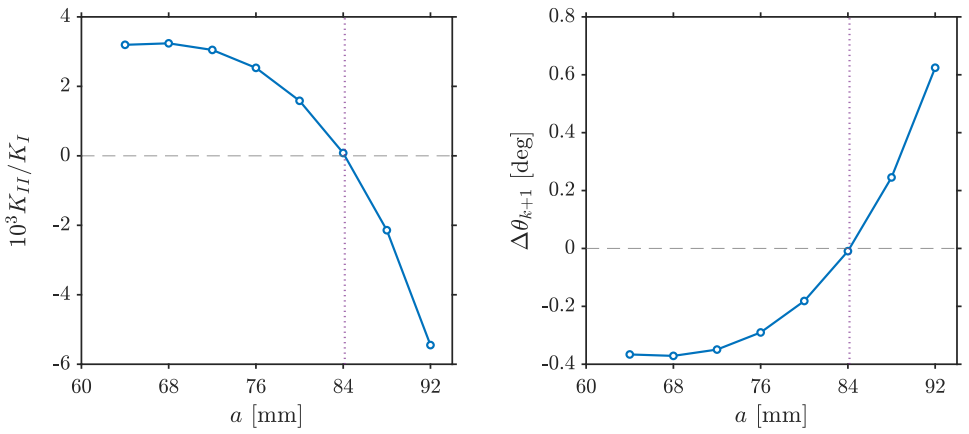
The sign change has an immediate directional consequence. At P_{21} the MTS turn toward P_{22} is still clockwise, whereas the negative mode mixity at P_{22} produces a counterclockwise turn toward the accepted P_{23} segment (Fig. 5). At that last solved state, $K_{\mathrm{I}} = 1.2444$, $K_{\mathrm{II}} = -0.0067803$, and $q_{\mathrm{K}} = -0.0054488$; the MTS rule then predicts $\Delta\theta_{24} = 0.62435°$ and $\theta_{24} = -2.7762°$. This final prediction specifies the qualified extension, not an accepted elastic solution at P_{24}.

5.4 COD comparison and extraction sensitivity

Because the sign reversal of mode mixity changes the MTS turning direction, its persistence under another SIF extraction procedure is particularly relevant. Each of the eight COD fits gives $q_{\mathrm{K}} > 0$ at P_{21} and $q_{\mathrm{K}} < 0$ at P_{22}, in agreement with the EDI sign-change bracket. The quantitative differences depend on the chosen fitting window and polynomial degree (Fig. 6). Across 176 fits, the greatest absolute difference between the COD- and EDI-based MTS turns

Table 2: Characteristic accepted states. SIFs have units MPa$\sqrt{\text{m}}$; angles are in degrees. The interpolated row contains only a graphical zero estimate, not a solved stress field or crack direction.

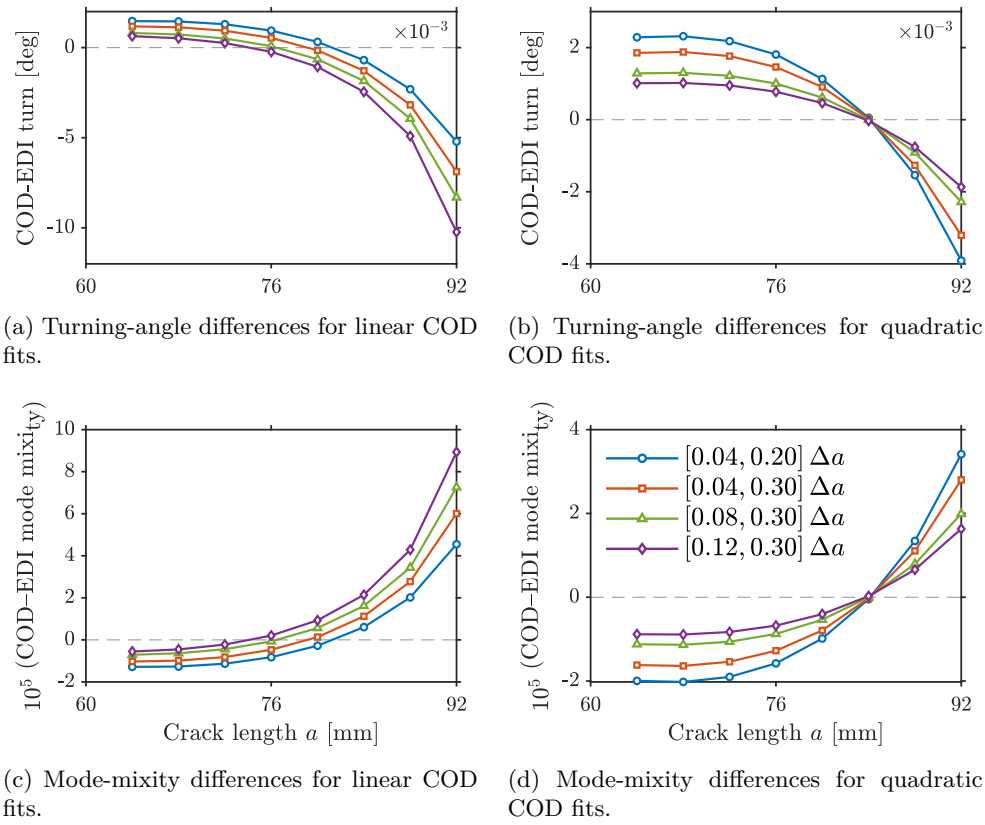
State	a [mm]	K_{I}	K_{II}	$K_{\text{II}}/K_{\text{I}}$	θ_k	$\Delta\theta_{k+1}$
P_1	4	0.36648	4.198×10^{-6}	1.146×10^{-5}	0.0000	-0.00131
P_{17}	68	0.78679	2.548×10^{-3}	3.239×10^{-3}	-2.4444	-0.37115
P_{21}	84	0.96631	7.860×10^{-5}	8.134×10^{-5}	-3.6366	-0.00932
P_{22}	88	1.0667	-2.284×10^{-3}	-2.142×10^{-3}	-3.6460	+0.24541
P_{23}	92	1.2444	-6.780×10^{-3}	-5.449×10^{-3}	-3.4006	+0.62435
Interpolated	84.15	—	—	0	—	—



(a) Late mode-mixity history.

(b) Incremental MTS turn.

Figure 5: Mode-mixity sign change and reversal of the MTS turn. The dotted line denotes the interpolated mode-mixity zero. The turn computed at P_{23} ($a = 92$ mm) predicts the proposed, physically unsolved next segment.



(a) Turning-angle differences for linear COD fits.

(b) Turning-angle differences for quadratic COD fits.

(c) Mode-mixity differences for linear COD fits.

(d) Mode-mixity differences for quadratic COD fits.

Figure 6: COD–EDI differences along the late crack path for four normalized COD fitting windows and linear or quadratic extrapolation. Upper panels compare MTS turns; lower panels compare mode-mixity ratios. Zero denotes agreement of the extraction methods, not a known exact solution.

is 0.01024°, at P_{23} for the linear fit over $[0.12, 0.30]\Delta a$. The quadratic COD turns at that state range from 0.6204° to 0.6225°, compared with the EDI prediction of 0.62435°.

The distinction between the sign and magnitude of a small mode-mixity ratio is important close to local symmetry. At P_{21}, the rear-window linear fit exceeds the EDI ratio by 26.3%, although the corresponding turning-angle difference is only 0.00245°. The relative discrepancy is consequently sensitive to the small denominator in the comparison. The two extraction methods thus agree on the sign-change bracket, although their common displacement solution precludes an independent estimate of the finite-element error. The differences among COD fits measure sensitivity to the extrapolation procedure, not statistical uncertainty.

5.5 Sensitivity to structured-tip resolution

For comparisons at matched crack lengths, δx and δy denote coordinate differences between the compared and reference paths, $\delta r = (\delta x^2 + \delta y^2)^{1/2}$ is their separation, and $\delta\theta$ is the difference of absolute directions in the frozen frame. The symbol $\Delta\theta_{k+1}$ remains reserved for a local MTS turn. Differences in mode mixity and interpolated crossing length are denoted by δq_{K} and δa_{LS}, respectively.

Two aspects of the characteristic mode-mixity history are examined on fixed accepted crack geometries. At the positive maximum P_{17}, reducing the structured-tip spacing from 0.027012 to 0.013506 mm changes K_{I} by 0.00134%, q_{K} by 1.05×10^{-6}, and the MTS turn by -1.20×10^{-4} degrees. The small differences arise from alternative discretizations of the same physical geometry.

The late sign-change bracket provides a more sensitive test. Table 3 compares the P_{21} and P_{22} fields at three near-tip resolutions spanning a factor of four in h_{tip}. For every tested level, q_{K} remains positive at P_{21} and negative at P_{22}. The bracket and the corresponding change in MTS

Table 3: Structured-tip sensitivity at fixed P_{21} and P_{22} geometries, with common interaction-domain radii and requested exterior sizing. The reported a_{LS} values are interpolated sign-change estimates, not independently solved zero-mode-II states.

Resolution	h_{tip} [mm]	$q_{\mathrm{K}}(P_{21})$	$\Delta\theta_{22}$ [deg]	$q_{\mathrm{K}}(P_{22})$	$\Delta\theta_{23}$ [deg]	a_{LS} [mm]
$2h_0$	0.054025	$+7.7957\times10^{-5}$	-0.008933	-2.1451×10^{-3}	$+0.245805$	84.14027
h_0	0.027012	$+8.1343\times10^{-5}$	-0.009321	-2.1416×10^{-3}	$+0.245406$	84.14637
$h_0/2$	0.013506	$+8.3031\times10^{-5}$	-0.009515	-2.1399×10^{-3}	$+0.245207$	84.14941

turning sense thus persist under the tested near-tip refinement.

The linearly interpolated estimate a_{LS} varies from 84.14027 mm at $2h_0$ through 84.14637 mm at h_0 to 84.14941 mm at $h_0/2$. Its successive shifts of 6.10 and 3.04 μm give an indicative contraction ratio corresponding to an apparent order of approximately 1.00. This figure describes the mesh sensitivity of an *interpolated estimate*, not convergence of a solved zero-mode-II point. Because the contributing physical states are separated by a 4-mm crack increment, the small shifts cannot be interpreted as micrometer-scale accuracy of the local-symmetry location.

Propagation with the $2h_0$ tip mesh produces a path nearly indistinguishable from the reference trajectory at specimen scale [Fig. 7a]. The maximum vertical-coordinate difference is 0.12760 μm and the maximum Euclidean separation of corresponding tips is 0.12800 μm, both at P_{23}. The maximum absolute direction difference is 0.24539 millidegrees at P_{22} [Figs. 7b and 7c].

The two independently generated paths also retain the positive mode-mixity maximum at P_{17} and the sign-change bracket between P_{21} and P_{22} [Fig. 7d]. The largest difference in q_{K} is 2.65×10^{-7}. Linear interpolation of the two independently propagated P_{21}–P_{22} brackets gives $a_{\mathrm{LS}} = 84.14637$ mm for h_0 and 84.146276 mm for $2h_0$, a difference of -0.000094 mm (-0.094 μm). As above, this is only a sub-increment sensitivity diagnostic, not a micrometer-accurate physical location of local symmetry.

The positive mode-mixity maximum, sign-change bracket, and turning reversal persist across the tested structured-tip resolutions, both on fixed geometries and along the propagated $2h_0$ trajectory. This evidence concerns the stated mesh perturbations; the conforming exterior connectivity also changes when the core is remeshed.

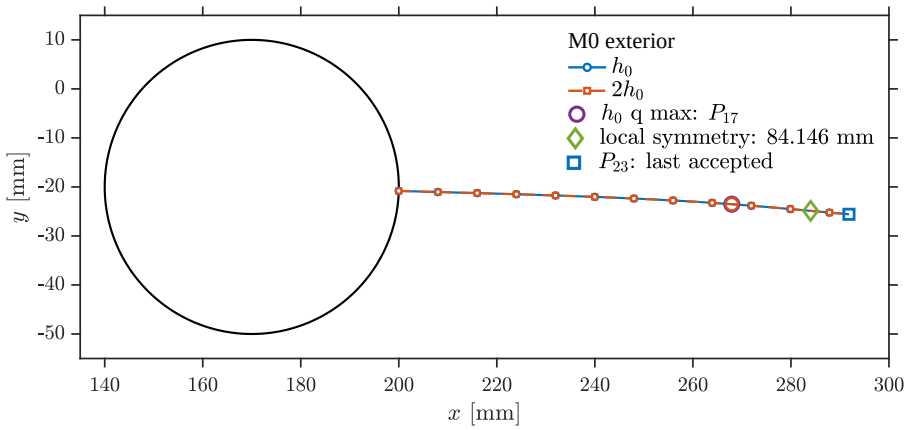
5.6 Sensitivity to exterior-mesh density

Coarsening the exterior mesh has little effect on the SIFs of the fixed P_{17} and P_{22} geometries. We therefore examined its influence on the complete trajectory by propagating the M1 discretization from the common first segment P_0P_1. All later crack directions follow from the M1 solutions rather than from the reference path.

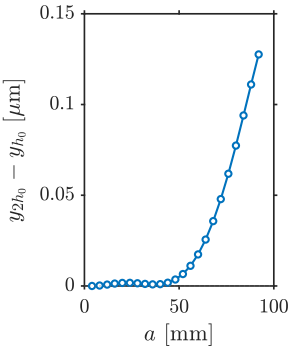
The two trajectories are nearly coincident at specimen scale [Fig. 8a]. Their small separation becomes apparent in the coordinate and angular differences in Fig. 8. The maximum vertical-coordinate difference is 0.06734 μm, the maximum Euclidean separation of corresponding tips is 0.06742 μm, and the maximum absolute direction difference is 0.07304 millidegrees, or 7.30×10^{-5} degrees [Figs. 8b and 8c].

The event sequence is also preserved when the exterior mesh is coarsened. Both propagated trajectories attain their positive mode-mixity maximum at P_{17} ($a = 68$ mm), with a positive ratio at P_{21} and a negative ratio at P_{22} [Fig. 8d]. Their largest absolute difference in q_{K} is 2.89×10^{-7}. Linear interpolation of the respective P_{21}–P_{22} brackets gives $a_{\mathrm{LS}} = 84.14637$ mm for the reference trajectory and 84.146 mm for M1, a difference of 0.000322 mm (0.322 μm). This sub-increment difference is reported only as a sensitivity diagnostic; neither interpolated value has physical precision at the micrometer scale.

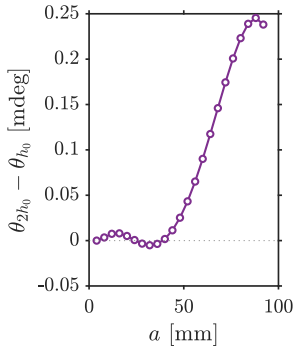
The characteristic event sequence is therefore insensitive to the tested coarsening of the *exterior* mesh. Because the near-tip discretization and interaction-domain radii are unchanged, the



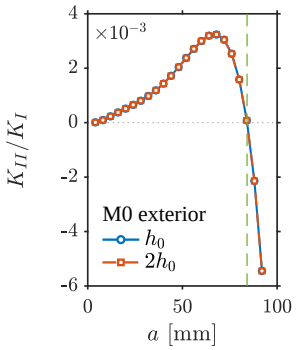
(a) Trajectories with structured-tip spacings h_0 and $2h_0$.



(b) Accumulated vertical tip-coordinate difference, $y_{2h_0} - y_{h_0}$.

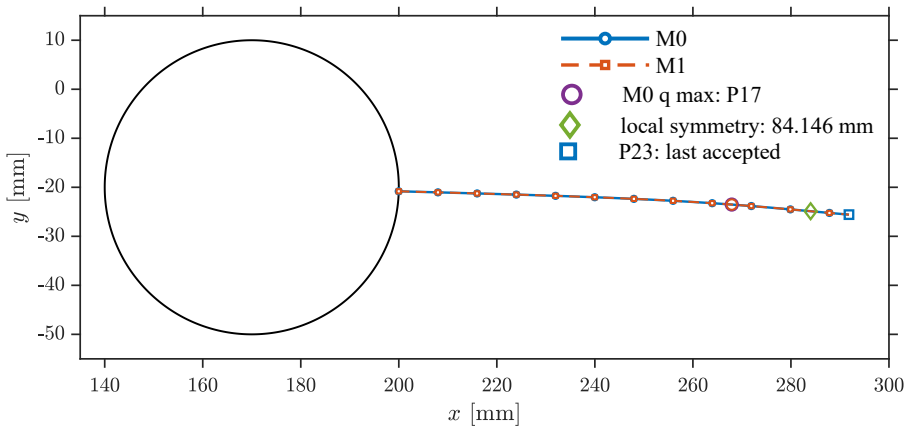


(c) Difference in the absolute crack direction in the frozen initiation frame, $\theta_{2h_0} - \theta_{h_0}$.

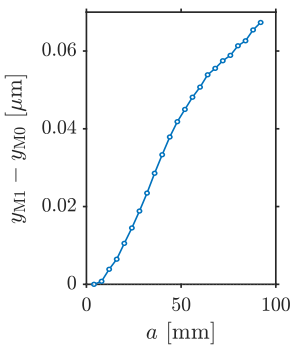


(d) Mode-mixity ratio $K_{\mathrm{II}}/K_{\mathrm{I}}$ along the two independently propagated trajectories.

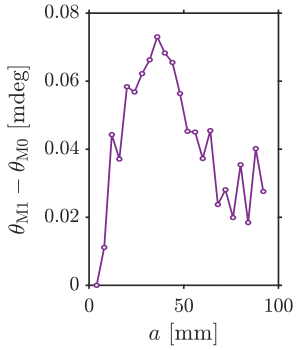
Figure 7: Effect of structured-tip resolution on the propagated trajectory. The h_0 and $2h_0$ paths share the interaction-domain radii and prescribed exterior sizing. The diamond indicates the interpolated mode-mixity zero of the reference path.



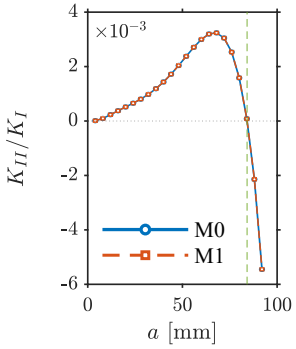
(a) Crack trajectories with the M0 and M1 exterior sizing laws.



(b) Accumulated vertical tip-coordinate difference, $y_{M1} - y_{M0}$.



(c) Difference in the absolute crack direction in the frozen initiation frame, $\theta_{M1} - \theta_{M0}$.



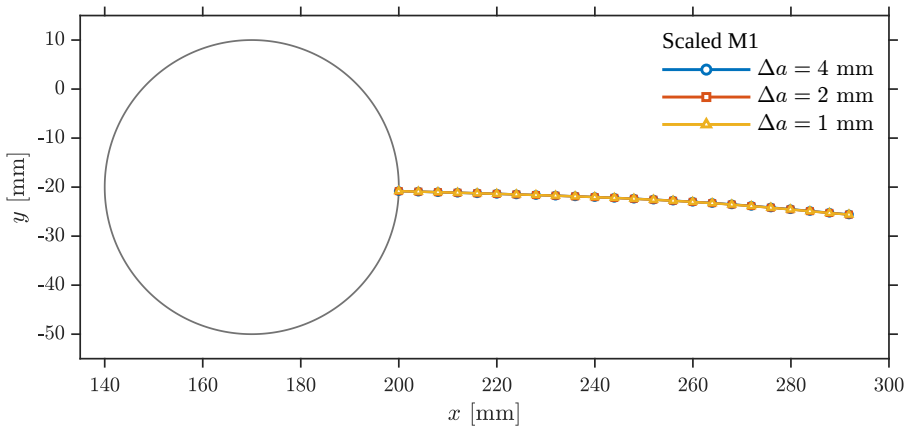
(d) Mode-mixity ratio K_{II}/K_I along the two independently propagated trajectories.

Figure 8: Effect of exterior-mesh coarsening on the crack path. The M0 and M1 trajectories are propagated separately with identical near-tip discretizations and interaction-domain radii. The diamond marks the interpolated mode-mixity zero of the reference path.

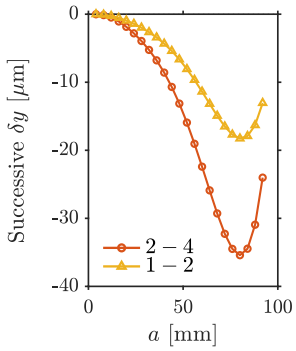
comparison does not establish general mesh independence.

5.7 Coupled crack-increment and discretization refinement

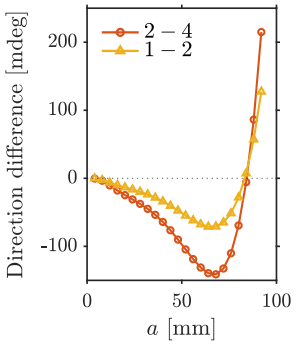
The paths computed with $\Delta a = 4$, 2, and 1 mm remain nearly coincident through 92 mm of growth [Fig. 9a]. Their differences in tip position and segment direction nevertheless decrease with refinement [Fig. 9]. In this study, the near-tip mesh, interaction domain, and exterior scales are reduced together with Δa, as specified in Section 4; hence the comparison concerns a coupled increment/discretization family.



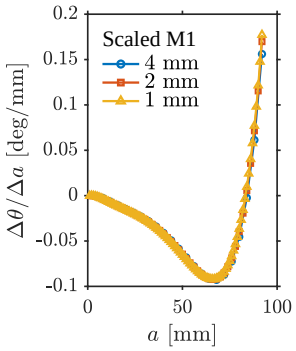
(a) Independently propagated crack trajectories for $\Delta a = 4$, 2, and 1 mm, all continued to 92 mm.



(b) Successive vertical-coordinate differences, $y_{2\,\mathrm{mm}} - y_{4\,\mathrm{mm}}$ and $y_{1\,\mathrm{mm}} - y_{2\,\mathrm{mm}}$, evaluated at exact common crack lengths.



(c) Successive differences in absolute crack direction in the frozen initiation frame.



(d) Turning density $\Delta\theta/\Delta a$ along the three independently propagated trajectories.

Figure 9: Coupled crack-increment and discretization refinement. The 4-, 2-, and 1-mm paths share dimensionless mesh and interaction-domain scales. Panels (b,c) compare paths at common crack lengths; panel (d) shows turning density on each path's native grid.

Table 4 compares tip positions and directions at crack lengths common to all three paths. The maximum tip separation contracts from 35.472 μm for 4 versus 2 mm to 18.291 μm for 2 versus 1 mm. The corresponding vertical differences decrease from 35.418 to 18.263 μm, while the maximum absolute direction difference decreases from 214.931 to 127.391 millidegrees.

Table 4: Successive differences between independently propagated paths in the dimensionless crack-increment refinement family. Comparisons use exact common crack lengths; no path interpolation is used.

Comparison	$\max \lvert\delta y\rvert$ [μm]	$\max \delta r$ [μm]	$\max \lvert\delta\theta\rvert$ [mdeg]	$\max \lvert\delta q_{\mathrm{K}}\rvert$
2 mm − 4 mm	35.4181	35.4721	214.931	2.48578×10^{-3}
1 mm − 2 mm	18.2629	18.2911	127.391	1.41618×10^{-3}

The contraction ratios are 1.939 for maximum tip separation and 1.687 for maximum direction difference. If d denotes a successive-difference maximum, the descriptive quantity $p = \log_2(d_{\Delta a}/d_{\Delta a/2})$ takes values of approximately 0.96 and 0.75, respectively. These are not formal Richardson orders: the compared paths result from recursively updated geometries, the maximizers need not occur at the same crack length, and both the increment and discretization scales change. These values describe contraction within the three-level family and cannot be regarded as measured asymptotic rates.

The normalized mode-mixity response exhibits a complementary trend. As shown in Table 5, the positive maximum of q_{K} falls from 3.23937×10^{-3} at 4 mm to 1.61266×10^{-3} at 2 mm and 8.04068×10^{-4} at 1 mm. Despite this nearly proportional decrease, the maximum divided by Δa changes by only about 0.71% between the coarsest and finest levels. The corresponding MTS turning density ranges from -0.092799 to -0.092139 deg/mm. The turning-density histories in Fig. 9d exhibit similar shapes on their respective crack-length grids, although this does not establish convergence of a smooth limiting curvature.

The positive maximum is sampled at $a = 68$ mm on the 4-mm path and at $a = 66$ mm on the two finer paths; the values therefore do not correspond to an identical crack length. Nevertheless, their near-proportional decrease with Δa, together with the stability of turning density, is consistent with the conditional curvature–MTS correspondence in Eq. (17). The observed behavior may be summarized as

$$q_{\mathrm{K}} = O(\Delta a), \qquad \Delta\theta = O(\Delta a), \tag{28}$$

within the investigated numerical family. Three coupled refinement levels provide evidence of this scaling rather than an independent proof of asymptotic convergence.

The estimated position of the mode-mixity sign change also shifts less with successive refinement. Linear interpolation gives $a_{\mathrm{LS}} = 84.148913$, 83.550120, and 83.268112 mm for the three levels. The successive differences are 0.59879 and 0.28201 mm, giving a contraction ratio of 2.12 and a descriptive apparent order near 1.09. On the 1-mm path, the sign change is bracketed by the solved states at 83 and 84 mm. None of these interpolated estimates is itself a solved zero-K_{II} state.

The nearly proportional decrease of the raw mode-mixity maxima and the stability of the increment-normalized measures accord with the conditional MTS–curvature relation in Eq. (17).

Table 5: Characteristic quantities in the crack-increment refinement family. The local-symmetry location is obtained by linear interpolation between the two solved states bracketing $q_{\mathrm{K}} = 0$.

Δa [mm]	$a(q_{\mathrm{K,max}})$ [mm]	$q_{\mathrm{K,max}}$	$q_{\mathrm{K,max}}/\Delta a$ [mm^{-1}]	$(\Delta\theta/\Delta a)_{q_{\max}}$ [deg/mm]	a_{LS} [mm]
4	68	3.2394×10^{-3}	8.0984×10^{-4}	−0.092799	84.148913
2	66	1.6127×10^{-3}	8.0633×10^{-4}	−0.092398	83.550120
1	66	8.0407×10^{-4}	8.0407×10^{-4}	−0.092139	83.268112

The available three-level evidence does not establish a unique smooth limiting trajectory.

6 Discussion

6.1 Mechanical interpretation of the turning reversal

The turning sense of each new segment follows from the current-tip mode-mixity ratio through the MTS criterion. Positive q_{K} produces a clockwise correction under the adopted sign convention; when q_{K} becomes negative, the correction is counterclockwise. The positive mode-mixity maximum at P_{17}, followed by the sign change between P_{21} and P_{22}, therefore explains the changes in turning direction along the accepted trajectory. At the same time, K_{I} increases under the fixed unit remote traction. The reversal is consequently a change in the relative sliding-mode contribution, not a decrease in the opening-mode intensity or a reversal of the crack's vertical displacement.

The remaining horizontal ligament is only 8.16 mm at P_{23}, suggesting that the approaching traction-free right boundary may contribute to the late mixed-mode response. That influence cannot, however, be identified separately from the increasing crack length and the evolving crack geometry in the present calculations. A controlled comparison at different boundary distances would be needed to determine its contribution. The computed SIFs establish the MTS turning mechanism, but not a unique physical cause for their evolution.

6.2 Numerical sensitivity and the smooth-path interpretation

The verification studies address different sources of uncertainty. Prescribed asymptotic fields test the implementation of SIF extraction, whereas COD fitting and the interaction integral provide two estimates derived from the same finite-element displacement field. Their agreement near the mode-mixity sign change lends support to the extracted turning sense, but cannot serve as an independent validation of the underlying elastic solution. Likewise, the centered-hole test establishes the expected reflection symmetry and a small one-step angular response, without implying attraction of an entire evolving trajectory.

The independently propagated mesh studies examine whether discretization differences alter the recursive path construction. Coarsening the exterior mesh in M1 preserves the paired near-tip core and interaction-domain support; changing the structured-tip spacing preserves the requested exterior sizing law, though conforming remeshing can change the exterior connectivity. Both propagated alternatives retain the positive mode-mixity maximum and the P_{21}–P_{22} sign-change bracket. Their maximum corresponding-tip separations, 0.0674 and 0.128 μm, respectively, are small relative to the crack increments. These observations demonstrate low sensitivity to the two tested mesh perturbations, not independence of the solution from all discretization choices.

The 4/2/1-mm trajectories probe a different aspect of the algorithm. In that family, the crack increment, structured-tip spacing, interaction-domain radii, and exterior mesh scales decrease together. Successive differences in tip position and direction contract, while the positive maximum of q_{K} decreases approximately in proportion to Δa. Its increment-normalized value and the corresponding turning density remain comparatively stable. The small-mixity MTS expansion thus provides a plausible connection to the signed curvature of a smooth limiting trajectory [Eq. (17)], provided such a limit exists and the discrete turning densities converge. Neither a formal convergence order nor the separate contribution of crack-increment error follows from these three coupled refinement levels. This distinction also limits direct comparison with studies that vary the increment alone, including the two-level comparison of Fang et al. [11].

Table 6 compares the exterior-mesh and structured-tip perturbations. The larger differences observed in the coupled 4/2/1-mm family do not isolate the contribution of any individual numerical scale.

Table 6: Trajectory sensitivity to exterior-mesh and structured-tip resolution. Differences in a_{LS} compare linearly interpolated mode-mixity zeros between P_{21} and P_{22}.

Perturbation	$\max\lvert\delta y\rvert$ [μm]	$\max\delta r$ [μm]	$\max\lvert\delta\theta\rvert$ [mdeg]	$\max\lvert\delta q_{\mathrm{K}}\rvert$	δa_{LS} [μm]
Coarser exterior M1	0.06734	0.06742	0.07304	2.89×10^{-7}	-0.322
Isolated-core $2h_0$	0.12760	0.12800	0.24539	2.65×10^{-7}	-0.094

Further tests should vary the interaction-domain radii while holding the physical crack geometry and other numerical parameters fixed. The existing mesh-quality, residual, and prescribed-field checks establish admissibility of the numerical calculations, but they do not provide a comprehensive error bound on the physical SIFs or trajectory. The earlier 8-mm diagnostic is retained as an extraction test and is not part of the coupled refinement sequence.

6.3 Last accepted state and physical scope

The accepted crack history ends at P_{23}. The proposed $P_{23}P_{24}$ extension satisfies geometrical and prescribed-field checks, but the attempted elastic solution for the 96-mm crack did not meet the PCG residual criterion. The resulting numerical stopping point cannot be interpreted as fracture arrest or contact with the right boundary.

The model treats one active crack in a homogeneous, linearly elastic plate and supplies only the directions of imposed quasi-static extensions. The stress-based initiation rule is not a subsequent propagation threshold, and the calculation does not determine growth loads, rates, or arrest. Nor does it model branching, competing initiation, plasticity, or dynamic fracture. These processes remain outside the formulation rather than being ruled out by the computed path.

7 Conclusions

An incremental LEFM procedure has been used to obtain the trajectory of a single crack emanating from an asymmetrically located circular hole. The uncracked-hole stress field selects the initiation point, and a prescribed outward-normal first segment establishes the initial crack geometry. All subsequent directions follow from current-tip interaction-integral SIFs and the MTS criterion, with the entire previous path retained in each elastic problem. The accepted 4-mm history extends to 92 mm; the next qualified geometry has no accepted physical solution.

The trajectory exhibits a characteristic change in its mixed-mode response. While K_{I} increases under the fixed reference traction, positive mode mixity reaches a maximum at P_{17} and subsequently changes sign between P_{21} and P_{22}. The associated reversal of the MTS turn reduces the negative inclination of the last accepted segment. The calculations identify this directional mechanism, without isolating the contribution of the approaching free boundary.

The numerical comparisons support the persistence of these characteristic events under the tested near-tip and exterior-mesh changes. A centered-hole control verifies the one-step response under reflection symmetry but does not establish stability of the evolving path. In the independently propagated 4/2/1-mm family, the geometric and directional differences contract as the increment and numerical length scales decrease together. The nearly proportional reduction in raw mode mixity and the relative stability of increment-normalized turning measures are consistent with a conditional smooth-path interpretation, not a proven continuum-convergence result.

The present findings concern LEFM direction selection for prescribed quasi-static increments, without a fracture-growth or arrest criterion. Interaction-domain sensitivity and controlled boundary-distance comparisons remain useful next steps for separating numerical extraction effects from the physical origin of late crack redirection.

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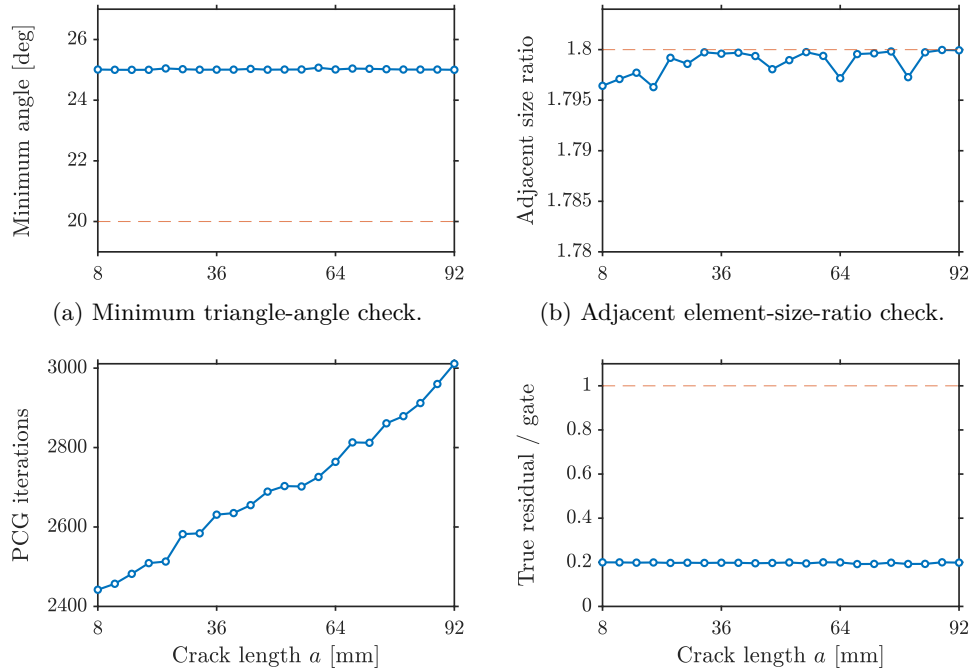
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A Supplementary numerical diagnostics

The accepted solver and mesh checks are summarized in Fig. S1. They provide reproducibility and acceptance diagnostics, rather than an estimate of the discretization error in the trajectory.



(a) Minimum triangle-angle check.

(b) Adjacent element-size-ratio check.

(c) PCG iterations for accepted physical solves.

(d) True relative residual normalized by its acceptance gate.

Figure S1: Diagnostics for accepted P_2–P_{23} states. Dashed lines are the 20° minimum-angle gate, 1.8 adjacent-size-ratio gate, and normalized true-residual gate of one. The separately enforced reported-PCG tolerance is 10^{-10}; the true-residual acceptance denominator is 5×10^{-10}. No failed P_{24} solve is plotted.